Does the galactic acceleration scale follow the cosmic age? A registered prediction, a0 ∝ 1/t, confronted with rotation curves at z = 1-2

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Version 2.42.1, 2026-10-13. Archived on Zenodo, https://doi.org/10.5281/zenodo.23329094. Licence CC BY 4.0. Developed with the help of AI assistants, as declared in the record.

Student version. This PDF is the student version of the record, written with an AI assistant from the archived text. The archived text remains on Zenodo: https://doi.org/10.5281/zenodo.23329094

Abstract

Version 2.42.1 (correction). The clock law tested here (a0 ∝ 1/t) was a translation introduced by the AI assistant, not a premise of the author, whose own reading of the distal scale is the Hubble radius c/H, i.e. a0 ∝ H(z). Section 5 of this record already found that, with a free amplitude, H describes the MUSE-DARK III bins better than the clock law (χ^2 5.1 against 14.9); with the local calibration fixed, the ordering reverses (χ^2 16.4 for the clock law and 25.2 for H). The follow-up record «Can one hidden-sector reading cover several dark-matter puzzles?» (DOI 10.5281/zenodo.23329080) adopts a0 ∝ H(z) as a phenomenological law chosen from these data and registers it for future data. CT1 stays as registered; every program, result and log is unchanged (new section 12 of the README).

Does the acceleration scale of galaxies follow the age of the universe? A registered prediction, a0 ∝ 1/t, confronted with distant galaxies

Student version of MM record 2.42.1 (DOI 10.5281/zenodo.23329094), written with an AI assistant from the archived text; the archived text remains on Zenodo.

The archived text is in English.

Ugo Lanciano, September-October 2026. Version: see the Zenodo record. Licence CC BY 4.0.

A correction to keep in mind from the start. Version 2.42.1 adds a section, section 12, with a correction. The clock law tested in this work is not a premise of the author. It is a *translation* made by the AI assistant, that is, a way of writing an idea of the author in formulas. The follow-up record uses another law, a0 ∝ H(z). It chose it by looking at the data of MUSE-DARK III and registered it for future data. All the programs, the results and the logs of the calculations of this work stay the same.

MM, that is, Mathematical Materiality, is the author's framework of ideas. The author has the idea of a *distal scale*: a distant scale, tied to the whole universe. This record tests the way in which the assistant translated that idea: tying it to the age of the universe. Section 12 clarifies this difference.

Before we start: the words we need

Galaxies rotate. A disc galaxy is an enormous disc of stars and gas that rotates. The stars and the gas turn around the centre. At each distance from the centre one can measure the velocity at which they turn. The graph of the velocity as a function of the distance from the centre is called the *rotation curve*.

What one would expect. In the Solar System distant planets turn more slowly than near ones. Neptune goes more slowly than the Earth. With Newton's law and only the matter we see, the distant stars of a galaxy should also turn more slowly. Instead, in galaxies the velocity stays high even far from the centre.

Acceleration. A body that turns in a circle has an acceleration towards the centre, even if it always goes at the same speed: the direction of its motion changes. Acceleration says how much the velocity changes each second, in size or in direction. It is measured in metres per second per second, m s^{-2}.

The radial acceleration relation. Astronomers compare two accelerations at each point of a galaxy. The first is the *observed* one, g_obs, obtained from the motion of the stars and the gas. The second is the one that ordinary matter alone would give with Newton's law, g_bar. The link between the two is called the *radial acceleration relation*. This link has a single scale, called **a0**.

How much a0 is. $a0 \approx 1.2 \times 10^{-10}$ m s^{-2}. It is a very small number. An example with numbers chosen by us: with this acceleration, an object at rest would take about 260 years to reach the speed of one metre per second. Where the acceleration is much larger than a0, Newton's law holds. Where it is smaller, gravity seems stronger than the matter we see would say.

Ordinary matter and dark matter. Ordinary matter is made of *baryons*: it is the matter of stars, of gas, of planets. When the observed attraction is larger than that of the baryons, the difference is attributed to *dark matter*.

Looking far is looking back. The light of a distant galaxy left a long time ago. While it travels, the universe expands and the light is stretched. Its wavelengths shift towards the red. This redshift is denoted by **z**. For example, z = 1 means that the wavelength doubled during the journey. The larger z is, the further back in time we look.

The age of the universe. In the calculations of this work today's age is t0 = 13.778 Gyr. Gyr means billions of years. When we look at a galaxy at z = 2, we see it as it was when the universe was much younger.

The dex. Many results are written in *dex*. A dex is a factor of ten, on a logarithmic scale. 0.3 dex is a factor of two. So *+0.3 dex* means about double, and *−0.3 dex* about half.

The χ^2 and the σ. The χ^2 (read *chi-squared*) is like an error score. It summarises how far a law departs from the data. The lower it is, the better the law follows the data. The σ (sigma) is the typical error of a measurement. To say that a value is *at 4 σ* means that it is four typical errors away from another.

In short

The question. In disc galaxies the radial acceleration relation has a single scale, $a0 \approx 1.2 \times 10^{-10}$ m s^{-2}. This number is close to $cH0/2\pi$ (Milgrom 1983). Here c is the speed of light and H0 says how fast the universe is expanding today. But does a0 change with cosmic time? It can be tested with the rotation curves of distant galaxies, at z ~ 1-2 (Milgrom 2015, 2017). The work MUSE-DARK III (Ciocan and colleagues, 2026) reports that a0 grows with redshift.

The law fixed in advance. This record fixed a law before making these comparisons. The data sets, however, were already known. The law is called **CT1**:

$a0(z)/a0(0) = t0/t(z)$.

In words: the value of a0 in an epoch, divided by today's, is equal to today's age divided by the age at that time. The younger the universe, the larger a0. An example without new numbers: if the universe had half of today's age, a0 would be double.

- The age t of the universe is computed from the expansion history of MM 2.32. All the 8896 samples of its calculation chain are used.
- The clock starts at the beginning of the model.
- CT1 predicts an a0 larger by 2.35 times at z = 1 and by 4.18 times at z = 2.
- The uncertainty that comes from cosmology is small. The 95 % band is ±0.42 % at z = 1 and ±0.49 % at z = 2.

An independent check. The same clock was also computed with standard cosmology, called ΛCDM. The difference is 0.14 % at z = 1 and 0.35 % at z = 2. Comparing sample by sample, it reaches up to 1.8 % at z = 2.5. So the test tests the clock, not the MM cosmology.

Four comparisons with published data. The first three, taken at face value, prefer a constant a0.

- **RC100.** These are 100 galaxies between z = 0.6 and z = 2.5. For those at $z \geq 2$ one looks at the dark-matter fraction at the *effective radius*, that is, the radius that contains half of the light. From the kinematics, with a

constant a0, the predicted fraction is 0.24. With CT1 it is 0.63. The published value is 0.26.
- **MUSE-DARK II.** These are 95 low-mass galaxies at z ≈ 1. For them the *baryonic Tully-Fisher relation* does not change: its zero point shifts by 0.00 ± 0.06 dex. (We see what the zero point is in section 3.) In the low-acceleration regime assumed here, CT1 instead predicts a shift of −0.26 to −0.32 dex.
- **KMOS3D.** These are galaxies with a lot of mass. Their zero point rises by 0.17 dex from z ≈ 0.9 to z ≈ 2.3. CT1 instead predicts a fall of about 0.2 dex.

The fourth comparison goes the other way. It is **MUSE-DARK III**: four groups of galaxies, called *bins*, between z = 0.5 and z = 1.3. The values come from a *digitization* of the published figure: someone read the values from the graph, point by point.

- Here a0 grows with redshift, and a constant a0 is the worst law.
- With the local calibration held fixed, CT1 has a better nominal statistic than a0 ∝ H.
- If instead the amplitude is left free, H describes the shape of the growth better.
- The measured growth is slower than that of CT1. Between the last and the first bin a0 grows by 1.36 times. CT1 predicts 1.75 times.

The data sets are therefore in tension with each other. One possible explanation is that they measure the masses in different ways. This record does not isolate this explanation.

What CT1 would need. To agree with the first comparisons, CT1 would need galaxies with less ordinary matter than that used.

- For RC100 at z ≥ 2, the mass shift that minimises the nominal χ^2 of CT1 corresponds to about half of the adopted baryonic mass.
- At z < 1.2 it corresponds to 0.78 times the mass.
- In the assumed regime, MUSE-DARK II requires a similar factor: from 0.48 to 0.55.
- At z ≥ 2, at the central values, the mass that CT1 requires is less than the mass of the stars alone for 26 RC100 galaxies out of 41.

What the measured gas says. The measurements of gas in galaxies that form stars, at z ≈ 1-2, go the other way.

- Molecular gas is measured from the lines of a molecule, carbon monoxide (CO), and from dust. These measurements are calibrated on 1444 galaxies, with three methods. At z ≈ 2 molecular gas is comparable to the stars, or larger.
- Atomic hydrogen at z ≈ 1 has on average the same mass as the stars.
- These are results on whole populations of galaxies. They are not measurements of the galaxies compared here.

The outcome. CT1, as registered, is in tension with the comparisons adopted. The tension depends on the mass models and on the simplifications declared. This outcome hits the reading CT1, that is, the cosmic-age clock: it does not test the MM framework (section 7).

1. The law tested (CT1)

The law. a0(z) = a0(0) · t0/t(z).

- t(z) is the age of the universe at the epoch that we see at redshift z. In a formula it is t(z) = ∫_z^∞ dz′/[(1 + z′) H(z′)].
- This age is called the *proper age of the comoving FLRW congruence*. In simple words: it is the time marked by a clock that moves together with the expansion of the universe, like a galaxy carried by the expansion.
- H is the expansion rate of the universe. Here H comes from MM 2.32.

The assumptions. They are the *bridge* between the idea and the numbers. They are not derived from the equations of 2.32.

1. **The clock of the medium is the cosmic age.** In a formula: U^μ∇_μT = 1, with T = 0 on the initial surface of the model. It means that T grows by one second every second, along the motion of matter, starting from the beginning of the model. Then T = t: the clock marks exactly the age of the universe. There is no different formation age for each galaxy, nor an offset.
2. **a0 = ξc/T.** Here ξ (read *xi*) is a number that is the same everywhere and always. It is obtained from today's local value, a0(0) = 1.2×10^{-10} m s^{-2}. This gives ξ = 0.174. It is a *calibration*, that is, a value adjusted on a measurement. It is not a prediction.

3. **The same radial law as always.** The value a0(z) enters the radial law already adopted in the programme. It is the "MLS" form of the radial acceleration relation: g_obs = g_bar/[1 − exp(−√(g_bar/a0))]. The formula says how to go from the acceleration of ordinary matter, g_bar, to the observed one, g_obs.
4. **The galaxies are in the right regime.** It is assumed that the galaxies tested are where that law holds. External fields, pressure and the selection of the galaxies are treated in the same way for every hypothesis.

The numbers. They are computed by the program `codice/ct1_previsione.py`. The 95 % band takes into account only the cosmological uncertainty. In square brackets there is the band; ΛCDM is the standard cosmology.

z	CT1, a0(z)/a0(0)	same clock in ΛCDM (independent check)	a0 ∝ H	baryonic Tully-Fisher shift, asymptotic (dex)
0.5	1.6025 [1.5975, 1.6081]	1.6024	1.325	−0.205
1.0	2.3476 [2.3378, 2.3574]	2.3508	1.787	−0.371
1.5	3.2127 [3.1978, 3.2274]	3.2211	2.356	−0.507
2.0	4.1810 [4.1606, 4.2014]	4.1956	3.009	−0.621
2.5	5.2409 [5.2143, 5.2676]	5.2622	3.734	−0.719

How to read a row. At z = 2 CT1 predicts an a0 4.1810 times today's. The band goes from 4.1606 to 4.2014. The same clock with ΛCDM gives 4.1956. The law a0 ∝ H gives 3.009. The last column says by how much the zero point of the baryonic Tully-Fisher relation would fall in the extreme case: −0.621 dex.

- Today's age is t0 = 13.778 Gyr. The band goes from 13.744 to 13.814 Gyr.
- The independent check uses a flat ΛCDM universe, with the median values of the chain: Ωm = 0.3066 and h = 0.6764. It uses no MM program. Ωm is the share of matter in the universe; h is today's expansion rate, written as a pure number.

The alternative laws. They are computed on the same cosmological background and with the same normalisation.

- **Constant a0.** a0 never changes.
- **a0 ∝ H.** a0 is proportional to the expansion rate H. This law was considered by Milgrom (2015).
- **Verlinde.** In the work of Verlinde (2016) the cosmological acceleration cH0 sets the scale. The MOND-like scale is a_M = cH0/6. That work, however, does not derive the replacement H0 → H(z), that is, it does not say that the scale follows H in time. And it discusses the cosmological limits of its approximation.

What a success would show, and what not. The reasoning is in the document `documenti/SHAPE_TEST_REPORT_astra_IT.md`.

- The law CT1 makes 1/a0 *affine* in the look-back time. The *look-back time* is how long ago the light left. *Affine* means that, in a graph, the points lie on a straight line.
- There are three exact *degeneracies*. A degeneracy is a case in which two different laws give the same numbers, and the data cannot separate them.
- A constant a0 satisfies the affine form with zero slope: the line is horizontal.
- If the zero of the clock is left free, the ratio tends to one and imitates a constant.
- In the *matter era*, when the universe was dominated by matter, 1/t and H are proportional. There the two laws cannot be told apart.
- With the zero of the clock fixed, CT1 has no freedom left.
- A success would support the idea "a0 follows the cosmic age". It would not support the MM cosmology in particular.
- The second assistant made a design study. It used five bins, the synthetic scheme of the pilot test. To separate CT1 from an alternative with two parameters, such as a0 ∝ E^γ, a precision of about 3.4 % per bin in ln a0 was needed. It is a result of that scheme. It is not a general rule.

2. Test 1: the dark-matter fraction at the effective radius (RC100)

The data. They come from Table 3 of Nestor Shachar and colleagues (2023). For each galaxy the table gives three things.

- Re, the effective radius, which contains half of the light.
- Vc(Re), the circular velocity at that radius.
- fDM(Re), the dark-matter fraction at that radius. It is the part of the square of the velocity that baryons do not account for.

An example without new numbers: if fDM is 0.26, about a quarter of the attraction needed for that rotation does not come from ordinary matter.

Where the mass comes from. The fraction comes from the authors' *dynamical decomposition*. The authors divide the motion of the galaxy between the part due to baryons and the part due to dark matter.

- In this decomposition the baryonic mass is a free parameter, adjusted on the data.
- The authors, however, start from a reference value, called a Gaussian *prior*. It is centred on two masses added together.
- The first is the mass of the stars. It is obtained from *SED fitting*, that is, from the light of the galaxy in the various colours.
- The second is the mass of the gas. It is obtained from the *scaling relations* of Tacconi and colleagues (2020): average formulas that tie the gas to other properties of the galaxy.
- The table lists the final value, the adjusted one.

A galaxy beyond the edge. The galaxy U3 25105 is at z = 2.52. It is just beyond the grid of the CT1 samples, which reaches z ≤ 2.5. In tests 1 and 3 its value is *extrapolated*, that is, extended beyond the edge.

- The extrapolated step from z = 2.50 to z = 2.52 is a factor 1.00829.
- The exact step of the same clock in ΛCDM is 1.00845.
- The difference is 0.016 %. It mixes two things: the offset between MM and ΛCDM, and the error of the extrapolation.
- The same extrapolation can be tried on the exact ΛCDM curve, on the same grid. It errs by −0.014 % for the clock and by −0.007 % for H.

The method.

- From the kinematics one obtains the observed acceleration: g_obs = Vc²/Re.
- For each hypothesis on a0 the MLS law is solved and g_bar is found.
- From here comes the predicted fraction: f_pred = 1 − g_bar/g_obs.
- The predicted fraction is compared with the published fDM.
- The uncertainty combines the error on fDM and the errors on Vc and on Re. The errors are treated as independent of each other.
- Then a common shift of log g_bar is tried, from −0.6 to +0.3 dex. It represents a systematic error in the baryonic masses, the same for all galaxies.
- The shift reported is the one that minimises the nominal χ^2. By itself it does not show that everything fits.

The results for the 41 galaxies at z ≥ 2. Their median z is 2.22.

hypothesis	predicted fraction (median)	published (median)	nominal χ^2	below −2σ	mass shift minimising χ^2
a0 constant	0.24	0.26	68.9	2	0.00 dex
a0 ∝ H	0.55	0.26	189.6	17	−0.21 dex
CT1	0.63	0.26	278.5	19	−0.32 dex

How to read it. With a constant a0 the predicted fraction, 0.24, is close to the published one, 0.26. The χ^2 is 68.9 and only 2 galaxies are below −2σ. With CT1 the predicted fraction is 0.63, the χ^2 rises to 278.5 and 19 galaxies are below −2σ. To do better, CT1 would need masses smaller by 0.32 dex.

The nearer galaxies. At z < 1.2 there are 32 galaxies. The χ^2 is 68.2 with a constant a0 and 101.8 with CT1. CT1 would require −0.11 dex.

The registered expectations. The four expectations written before the calculation (E1-E4) all came true.

A limit. The χ^2 per galaxy is 1.7-2.1 even with a constant a0. It can mean two things. The nominal errors may be too narrow. Or neglected covariances (errors tied to each other) and imperfections of the model count.

A precedent. This goes in the direction of Milgrom (2017). Milgrom looked at six galaxies of Genzel and colleagues (2017). He wrote that these data "all but exclude" $a0 \approx 4a0(0)$ at $z \sim 2$. In simple words: they almost exclude that a0 at z ~ 2 is four times today's.

3. Test 2: the baryonic Tully-Fisher relation (MUSE-DARK II, KMOS3D)

What it is. Galaxies with more mass rotate faster. The *baryonic Tully-Fisher relation* ties the mass of ordinary matter of a galaxy, Mbar, to its rotation velocity, V. On a graph with logarithmic scales the points lie almost on a straight line.

The zero point. It is the height of that line, at fixed slope. In technical terms it is the intercept of log Mbar against log V. If the zero point falls, at equal velocity the galaxy has less mass.

Why not RC100 again. A Tully-Fisher relation built with the same galaxies would repeat test 1.

The method.

- The MLS law is used, at fixed radius and velocity.
- One looks at a galaxy in the regime $y0 = g_bar/a0(0)$. The number y0 says how large the acceleration of ordinary matter is compared with today's a0.
- If a0 grows, $a0(z) = R(z)a0(0)$, the galaxy needs a baryonic mass k times the one it needs with a0(0).
- The shift of the zero point is then log10 k.
- In the *deep regime*, that of very low accelerations, the shift tends to −log10 R(z).
- In the *Newtonian regime*, that of high accelerations, it tends to zero.
- The sizes of the galaxies are held fixed.

MUSE-DARK II

Independent of RC100. The galaxies are small and slow.

The assumption on the regime. This record assumes that their velocities at 2Re, that is, at two effective radii, lie at low accelerations: $y0 \lesssim 0.3$. There the relation depends on a0 more than on the sizes. The shifts below hold for that regime.

The representative galaxy. The source describes a typical galaxy.

- Its mass of stars is $M^* = 10^9$ solar masses.
- Its rotation velocity is 70 km/s at 1.8Re.
- The velocity *dispersion* is 30 km/s. It is how much the velocities of the stars and the gas are scattered around the mean motion.
- Putting rotation and dispersion together, the circular velocity is about 86 km/s.
- This galaxy alone does not establish the regime. The source gives neither its radius nor how its baryons are distributed.

hypothesis	predicted shift (y0 0.05-0.3)	distance from 0.00 ± 0.06 (published fit)	adding a common ±0.20 dex (a scenario chosen here)
a0 constant	0.00	0 σ	0 σ
a0 ∝ H	−0.17 to −0.21	2.8-3.6 σ	0.8-1.0 σ
CT1	−0.26 to −0.32	4.3-5.3 σ	1.2-1.5 σ

How to read it. The published zero point does not shift: 0.00 ± 0.06 dex. CT1 predicts from −0.26 to −0.32 dex. This is at 4.3-5.3 σ from the published value.

The last column. The ±0.06 dex of the published fit already contain the ±0.20 dex that the authors give to the individual baryonic masses. The last column adds the same amount again, as an error common to all the masses. It is a scenario chosen by this record. It is not a way of calculating the errors indicated by the source. With this extra error, CT1 goes down to 1.2-1.5 σ.

What CT1 would need.

- The true baryonic mass would have to be from 0.48 to 0.55 times the estimated one.
- Keeping the stars fixed, only 7-20 % of the estimated gas would remain.
- The share of gas in the estimate, 56 %, is an assumption for a representative galaxy.
- The source gives two medians separately: molecular gas 34 % and atomic gas 22 %. The sum of two medians is not necessarily the median of the sum.

KMOS3D

Not independent. KMOS3D studies the same population of galaxies as RC100.

What is measured. The zero point rises by +0.17 ± 0.06 dex from z ≈ 0.9 to z ≈ 2.3.

What the laws predict, at fixed regime.

- CT1: a fall of 0.17-0.21 dex.
- a0 ∝ H: a fall of 0.13-0.17 dex.
- Constant a0: no change.

A limit of the model. The fixed-regime model does not reproduce the rise for any law. One possible explanation is an evolution of the structure of the galaxies: more baryons inside the radius where the velocity is measured.

The registered expectations of test 2

- **T1 did not come true.** In the assumed regime CT1 gives from −0.26 to −0.32 dex. The registered expectation said from −0.30 to −0.37.
- **T2-T4 came true.**
- **A calculation error in the protocol.** The protocol used a representative velocity of 61 km/s instead of 77.8 km/s. The error is declared. The regime remains an assumption, as said above.

4. Test 3: are the masses overestimated? Stars and directly measured gas

The question. CT1 would do better if the masses of the distant galaxies were smaller than those used. Are the masses then *overestimated*, that is, estimated too large? This test checks it.

The method.

- For each RC100 galaxy one computes the baryonic mass that a hypothesis requires. Radius and mass distribution are held fixed.
- The formula is Mbar_req = Mbar · (1 − f_pred)/(1 − fDM).
- This mass is compared with the mass of the stars alone, obtained from SED fitting. The mass of the stars does not depend on the gas.
- Test 3 reuses the RC100 data and the predicted fractions of test 1.
- The gas scaling relation used (Tacconi and colleagues 2018, Table 3b) is a reduced version. It omits the term that depends on size, 0.11 log(Re/Re,0).

41 galaxies at z ≥ 2	required baryonic mass minus stellar mass (median)	galaxies below their stellar mass	implied gas fraction (median)
a0 constant	+0.17 dex	7	33 %
a0 ∝ H	−0.04 dex	23	negative
CT1	−0.16 dex	26	negative

How to read it. A galaxy cannot have less ordinary matter than its stars: the stars are already ordinary matter. With a constant a0 the required mass is above the stars, and the rest can be gas: 33 %. With CT1 the required mass is below the stars for 26 galaxies out of 41. The implied gas fraction would come out negative: less than zero gas would be needed.

The nearer galaxies. At $z < 1.2$ CT1 requires +0.11 dex above the stars: the implied gas is 23 %. A constant a0 requires +0.22 dex: the implied gas is 39 %.

What the scaling relations say. At $z \geq 2$, in the median, they give a gas mass 1.32 times that of the stars. The dynamical masses of RC100 already lie 0.19 dex below the sum of the stars and of the gas of the scaling relations. And they lie 0.22 dex above the stars alone.

A limit. These comparisons use central values and fixed mass distributions. The uncertainties on the stars and on the dynamical masses are not propagated in the calculations.

The gas measured directly.

- **Molecular gas.** The scaling relations of molecular gas are fits to 1444 galaxies that form stars, between $z = 0$ and $z = 4$. The gas is measured with three independent methods: the CO lines, the light of the dust in the far infrared, the photometry of the dust at a wavelength of about 1 mm. Once their zero points are aligned, the three methods follow the same trends (Tacconi and colleagues 2018).
- **How much it grows.** The ratio of molecular gas to stars grows as $(1 + z)^{2.5}$. The further back in time one looks, the more gas there is compared with stars.
- **Atomic hydrogen.** At $z \approx 1$ it is measured by adding up the 21 cm signals of many galaxies that form stars. On average atomic hydrogen has the same mass as the stars (Chowdhury and colleagues 2020). An average obtained by adding up the signals, however, is not the median fraction inside the kinematic radius.
- **In these populations gas is a large part of the baryons:** about 35-60 %, depending on the population and the kind of gas.
- **These are not measurements of the galaxies compared here.**

The comparison with CT1.

- In the assumed regime, CT1 allows the gas to be only 8-21 % of the baryons of the representative galaxy of MUSE-DARK II. Be careful: this is a quantity different from the 7-20 % of the estimated gas seen before.
- For RC100 at $z \approx 2$, at the central values, CT1 gives in the median a negative amount of gas.
- A common error in the zero points of the gas measurements is possible, of 0.1-0.2 dex. It can come for example from the factor that converts CO into molecular hydrogen (H2), or from the ratio between dust and gas. Under these assumptions it would not be enough.

The registered expectations. The four registered expectations (G1-G4) all came true.

5. Test 4: the growth reported by MUSE-DARK III (z = 0.5-1.3)

The data.

- They are four bins of a0(z).
- They come from Table 1 of a paper by Alcock (MNRAS, DOI 10.1093/mnras/stag1485).
- That table is a digitization of Fig. 3 of Ciocan and colleagues (MUSE-DARK III).
- The bins summarise a dynamical analysis of the galaxies. They are not new measurements.
- The table does not say with which convention the error bars are given. Nor does it give the *covariance* of the bins, that is, how much their errors are tied to each other.
- The error bars were made symmetric.

The method. The protocol was written before the calculation, but after the published data were known.

- **The calibration on SPARC.** a0(0) is calibrated on SPARC alone, a collection of nearby galaxies: 149 galaxies and 3150 points. The MLS law is used. The ratios between mass and light of the stars are 0.5 for the discs and 0.7 for the central bulges. Each galaxy has the same weight.
- **The value found.** $a0(0) = 1.019 \times 10^{-10}$ m s^{-2}. Then this value is *frozen*: it is not changed any more.
- **Its uncertainty.** It is estimated with the *bootstrap*: the calculation is repeated many times, reshuffling the galaxies. The 95 % band goes from 0.90 to 1.16. Multiplying the masses of the stars (discs and bulges) by 0.8 gives 1.29; by 1.2 it gives 0.83. The gas stays fixed.
- **No adjustment on the distant data.** Each law predicts the four bins without any fit on the distant data.

- **A second diagnostic.** It leaves one amplitude free for each law. In this way it tests only the *shape* of the growth.
- **How the bars are read.** The program reads the bars of the bins as 1σ, with a diagonal observational covariance (errors of the bins not tied to each other). The χ^2 values of the table come from here.
- **The other scenarios.** The protocol also foresees other scenarios: bars read as 95 %, correlations between the bins, extra systematic errors. These are reported in the report of the second assistant. The program distributed here does not recompute them.

law	predicted a0 at z = 0.50 / 0.83 / 1.05 / 1.28	nominal statistic (4 bins)	shape χ^2 (3 degrees of freedom)	last bin / first bin
data	1.99 / 2.20 / 2.57 / 2.71	—	—	1.36
a0 constant	1.02 / 1.02 / 1.02 / 1.02	168.6	28.8	1.00
a0 ∝ H	1.35 / 1.64 / 1.87 / 2.13	25.2	5.1	1.57
CT1	1.64 / 2.11 / 2.47 / 2.86	16.4	14.9	1.75
1 + z (chosen after seeing the data)	1.53 / 1.86 / 2.09 / 2.32	11.8	3.3	1.51

The units. The values of a0 are in units of 10^{-10} m s^{-2}.

What the nominal statistic contains. The errors of the bins, the spread of the local calibration and the cosmological spread of the samples.

Two clarifications.

- At these redshifts, the same clock in ΛCDM differs from CT1 by at most 0.2 %.
- Tests 1-3 use $a0(0) = 1.2 \times 10^{-10}$ m s^{-2}. Test 4 calibrates its own value on SPARC: 1.019×10^{-10} m s^{-2}. The shape test does not depend on this difference.

The result.

- The MUSE-DARK III bins prefer an a0 that changes over a constant a0. The constant law is the worst.
- With the local calibration frozen, CT1 has a better nominal statistic than a0 ∝ H: 16.4 against 25.2.
- With a free amplitude, H describes the shape better: χ^2 5.1 against 14.9, with 3 degrees of freedom.
- The central growth of CT1 is steeper than that of the bins. CT1 is too low at z = 0.5 and too high at z = 1.3.
- This comparison does not give the *robust* support that was required. But it goes in a different direction from the first three.
- The law 1 + z fits best of all. But it was chosen after seeing the data, and it is only a term of comparison.

6. What the tests say together

The first three comparisons. Taken at face value, CT1 needs baryonic masses of the distant galaxies smaller than those used.

- About half for RC100 at z ≥ 2 and for MUSE-DARK II in the assumed regime.
- 0.78 times for RC100 at z < 1.2.
- These are the shifts that minimise the nominal statistics.
- At z ≥ 2, at the central values, CT1 would want fewer baryons than stars for 26 galaxies out of 41.
- A constant a0 describes RC100, and the simplified comparison with MUSE-DARK II, better than CT1.
- The fixed-regime model does not reproduce the rise of the KMOS3D zero point (+0.17 ± 0.06 dex) for any law. An evolution of the structure of the galaxies is one possible explanation.

The two MUSE-DARK papers do not measure the masses in the same way. They also differ in the sample, in the selection and in the observed quantity.

- Paper III uses a dynamical normalisation of the disc. It finds a growth of a0.
- Paper II uses the masses of the stars from SED fitting and the gas from the scaling relations. It finds no evolution of the baryonic Tully-Fisher relation.

- The way of normalising the masses is a possible explanation of the difference. It is documented in the papers. This record does not isolate it.

Test 4 shows the tension from the other side. With the MUSE-DARK III bins a0 grows, but more slowly than CT1.

- A constant a0 at z ≈ 2 (RC100) and a growth at z = 0.5-1.3 (MUSE-DARK III) are in tension.
- They are not, however, logically incompatible. A non-monotonic evolution (which first rises and then falls), different populations of galaxies or systematic errors could reconcile them.
- None of the four comparisons gives CT1 the robust support required in its registered form.

The gas. The measurements of gas in populations of galaxies that form stars confirm large gas fractions at z ≈ 1-2.

- They do not measure the galaxies compared here.
- The uncertainties on the stars and on the dynamical masses are not propagated.
- So the idea "the masses are overestimated" is not excluded. But it would have to hold against these population results.

7. What is not claimed

- **It is not claimed that a0 is constant.** The comparison depends on the published masses and decompositions.
- **It is not claimed that CT1 is formally excluded.** The comparisons depend on the published mass models. And no *joint likelihood* with the covariances has been built, that is, a single calculation that puts together all the data with their tied errors. In its registered form, however, at the central values of the RC100 masses, CT1 would require fewer baryons than stars at z ≥ 2 for 26 galaxies out of 41.
- **It is not claimed that MM is tested.** Only the clock law is tested. The MM expansion history changes the central prediction by only 0.35 % at z = 2 (independent check of section 1).
- **It is not claimed that the MUSE-DARK III bins are independent measurements.** They are a digitization of a published figure. Test 4 was written after they were known. The law 1 + z was chosen after seeing them.
- **It is not claimed that the tests are independent of each other.**
- KMOS3D and RC100 have the same population of galaxies.
- Test 3 reuses test 1.
- Tests 1-3 depend on the gas scaling relations: the RC100 priors and the gas of MUSE-DARK II.
- Test 4 depends on the calibration on SPARC and on the dynamical normalisation of MUSE-DARK III.
- **It is not claimed that a0(0) or ξ is derived.** Both are calibrated. No galaxy data change R(z). The normalisation is calibrated on nearby galaxies, and the fits on the distant data serve only as diagnostics.
- **It is not claimed that a different clock would do better.** A clock started later would be a new hypothesis. It would have to be written before independent data, not adjusted on these.

12. The correction (added in version 2.42.1, October 2026)

What changes.

- In section 1 assumptions 1-2 tie the acceleration scale to a clock that reads the cosmic age.
- This link was a translation introduced by the AI assistant. It is not a premise of the author.
- The author's reading of the distal scale is the *Hubble radius*, R_H. It is the distance at which the expansion of the universe carries galaxies away at the speed of light.
- Suppose that a0 is proportional to c^2/R_H, with a factor that does not depend on redshift.
- If one identifies R_H = c/H, one gets $a0 \propto H(z)$. One does not get $a0 \propto 1/t$.

What the data of this work already said. Section 5 reports two results.

- With a free amplitude, the H law describes the MUSE-DARK III bins better than CT1: χ^2 5.1 against 14.9, with 3 degrees of freedom.
- With the local calibration frozen, CT1 was nominally better: 16.4 against 25.2.

The follow-up.

- The follow-up record is titled *Can one hidden-sector reading cover several dark-matter puzzles?* (Zenodo, October 2026). It can be found among the related works of this record.
- It repeats the comparison with Ωm = 0.315. It gets χ^2 5.5 for H against 15.3 for the age law.
- It adopts $a0(z) = A \cdot H(z)/H0$ as a *phenomenological* law, that is, one chosen because it describes these data. It is

not a confirmed law.
- It registers it for future data. The shape of the law is tested separately from the normalisation. A is free for each homogeneous sample of galaxies. The local value is treated as a separate test.

What does not change. CT1 stays as it was registered here, with its four comparisons and the tensions of section 6. Every program, result and log of the calculations of this work is identical to version 2.42.

How the work was done

The working names. In the working records (`records/`, older logs and notes) this deposit is called 2.40: it was its provisional name. The environment variable MM240_SPARC_DIR keeps that name.

Who did what. The author is not a professional physicist. The framework and its premises are his. The calculations were done with AI assistants.

- The CT1 samples, the shape test and test 4 were done by the second assistant (Codex, "Astra").
- The first three tests were done by Claude.
- Test 4 was then recomputed by Claude with a different code. It is the one in the folder `codice/`. The numbers agree (`records/NOTA_ESITO_CONTROLLO_MUSE_DARK3.md`).

The registrations. Each protocol was written before its calculation, and its digital fingerprint (SHA-256) was computed before the calculation (`records/`). The fingerprints do not have a public time stamp.

What is declared. One registered expectation (T1) did not come true and one protocol contains an arithmetic error: both are reported.

The protocols and the notes of the tests.

- Test 1: program `codice/controllo_z2.py`, protocol `records/PROTOCOLLO_CONTROLLO_Z2.md`, data `dati/rc100_table3.csv`. The same program prints the errors of the extrapolation for the galaxy U3 25105.
- Test 2: program `codice/controllo_tully_fisher.py`, protocol `records/PROTOCOLLO_CONTROLLO_TF.md`. The arithmetic error of the protocol is declared in `records/NOTA_ESITO_CONTROLLO_TF.md`.
- Test 3: program `codice/controllo_gas.py`, protocol `records/PROTOCOLLO_CONTROLLO_GAS.md`.
- Test 4: program `codice/controllo_muse_dark3.py`, protocol `records/PROTOCOL_CLOCK_TEST_astra.md`, data `dati/muse_dark3_bins_alcock.csv`. The other scenarios of the protocol are in the report `documenti/CLOCK_TEST_REPORT_astra_IT.md`.
- The CT1 prediction: program `codice/ct1_previsione.py`. The shape study: `documenti/SHAPE_TEST_REPORT_astra_IT.md`.

The review before publication. The second assistant reviewed the draft before publication. Its reports, in two rounds, are in `documenti/REVIEW_astra_20261011_IT.md` and `documenti/REVIEW_astra_20261011_round2_IT.md`. The corrections are listed in `documenti/CORRECTIONS_before_publication.md`.

The reference outputs. They were regenerated with Python 3.13.13, numpy 2.4.6, scipy 1.18.0 and matplotlib 3.11.1. The numbers stayed the same; some printed labels were corrected.

A corrected sentence. A sentence of the earlier working notes, on the origin of the RC100 masses, was inexact. It is corrected in `records/NOTA_ESITO_CONTROLLO_GAS.md`.

No human referee has checked this record.

How to redo the calculations (Python 3 with numpy, scipy and matplotlib).

```
python codice/run_all.py        # the five programs; outputs in uscite/, compared with risultati/
(about one minute);
                                # exit code 0 only if every log and JSON is reproduced identically
python codice/ct1_previsione.py
python codice/controllo_z2.py
python codice/controllo_tully_fisher.py
python codice/controllo_gas.py
python codice/controllo_muse_dark3.py
```

- Test 4 needs two files of the SPARC collection (Lelli, McGaugh and Schombert 2016), which are not

redistributed here: `SPARC_Lelli2016c.mrt` and `Rotmod_LTG.zip`, from http://astroweb.case.edu/SPARC/.
- They must be put in `dati/esterni/`, or in the folder indicated by the environment variable MM240_SPARC_DIR.
- The program checks their SHA-256 fingerprints, listed in `dati/README.md`.

The historical program. `documenti/calcola_ct1_astra.py` is the program that produced `dati/ct1_campioni.npz`. It is included as a historical source, but it does not run as distributed.

- It needs the reproduction kit of MM 2.32 (Zenodo, DOI 10.5281/zenodo.22840735) and CAMB 2.0.4.
- Its paths are those of the author's computer.
- It reads its protocol as `PROTOCOLLO_CT1.md`, next to itself. Here the protocol is distributed as `records/PROTOCOL_CT1_astra.md`.
- Without that file it writes the samples and then stops while it builds its summary.
- The comments of the code and the printed labels are partly in Italian.

The SHA-256 fingerprints of all the files are in `IMPRONTE.sha256`.

The Italian summary. The archived text ends with a *Sintesi in italiano* (a summary in Italian) of seven points. Its contents are all already in the sections of this version.

This version. This student version was written with an AI assistant (Claude) from the archived text in English. First a full Italian translation was made. Then the translation was rewritten for a school student. The numbers were checked with a program: the translation against the English text, this version against the translation. The archived text is the authoritative one.

This English text is a translation of the Italian student version, which the author read and approved.

Glossary

- **a0.** The acceleration scale of the radial acceleration relation. Today it is about 1.2×10^{-10} m s^{-2}.
- **Acceleration.** How much the velocity changes each second, in size or in direction. It is measured in m s^{-2}.
- **Baryonic Tully-Fisher relation.** The link between the mass of ordinary matter of a galaxy and its rotation velocity.
- **Baryons.** Ordinary matter: stars, gas, planets.
- **Bin.** A group of galaxies at similar distances, summarised by a single value.
- **Bootstrap.** A way of estimating an uncertainty: the calculation is repeated many times, reshuffling the data.
- **Calibration.** A value adjusted on a measurement. It is not a prediction.
- **χ^2 (chi-squared).** An error score: it summarises how far a law departs from the data. The lower it is, the better.
- **CT1.** The law a0 $\propto$ 1/t, with the clock started at the beginning of the model.
- **Dark matter.** The name given to what produces the extra attraction that baryons do not account for.
- **Deep regime and Newtonian regime.** The deep regime is that of accelerations much smaller than a0. The Newtonian one is that of much larger accelerations, where Newton's law holds.
- **Degeneracy.** A case in which two different laws give the same numbers, and the data cannot separate them.
- **Dex.** A factor of ten on a logarithmic scale: 0.3 dex is a factor of two.
- **Digitization.** Reading the values from a published graph, point by point.
- **Effective radius (Re).** The radius that contains half of the light of the galaxy.
- **fDM(Re).** The fraction of the square of the circular velocity at Re that baryons do not account for.
- **Fit.** The calculation that chooses the values of a formula so that it follows the data as well as possible.
- **H, H0.** The expansion rate of the universe. H0 is its value today; H(z) that of a past epoch.
- **Hubble radius.** R_H = c/H: the distance at which the expansion carries galaxies away at the speed of light.
- **ΛCDM.** The standard cosmology, with constant dark energy and cold dark matter.
- **MLS law.** g_obs = g_bar/[1 − exp(−√(g_bar/a0))]: the form of the radial acceleration relation used by the programme.
- **Prior.** The starting value given to a parameter before fitting it to the data.
- **Radial acceleration relation.** The link, point by point, between the observed acceleration and the one that ordinary matter alone would give.
- **Redshift (z).** How much the light has been stretched during its journey. The larger z is, the further back in time we look.
- **Rotation curve.** The graph of the velocity of the stars and the gas as a function of the distance from the centre of the galaxy.
- **Scaling relations.** Average formulas that tie a quantity, like the gas, to other properties of the galaxy.
- **SED fitting.** The way of obtaining the mass of the stars from the light of a galaxy in the various colours.
- **σ (sigma).** The typical error of a measurement.

- **Velocity dispersion.** How much the velocities of the stars and the gas are scattered around the mean motion.
- **Zero point.** The height of the Tully-Fisher line, at fixed slope. Technically: the intercept of log Mbar against log V.

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